

Physics-verified periorbital wearable for joint diabetic retinopathy and neuropathy screening

Analysis notebook: datasets, models, digital twin and results

Structure follows the project's analysis protocol: PART 1 integrity checks, PART 2 cohort structure, PARTS 3 to 5 the fundus grader and its transfer across populations, PART 6 the macular edema head, PART 7 the systemic (EHR) models with the leakage audit, PART 8 the physiological branch, PART 9 the digital twin credibility numbers, PART 10 design verdicts and the audit summary.

Inputs: `results/RESULTS_MASTER.json` (single source of numbers, with the source file and key path of every entry) and the stored per-subject predictions in `results/out/*.npz`. Training is done by the scripts in `results/src/` (see README). Outputs of this notebook (CSV tables and PNG figures) are written to `results/notebooks/out/`.

```
In [1]: import warnings; warnings.filterwarnings("ignore")
import json, pathlib, numpy as np, pandas as pd, matplotlib.pyplot as plt
from sklearn.metrics import (roc_auc_score, roc_curve, cohen_kappa_score, confusion
                             average_precision_score)
from sklearn.linear_model import LogisticRegression
from sklearn.calibration import calibration_curve
from scipy.stats import fisher_exact, spearmanr, mannwhitneyu

RS = 20260828
ROOT = pathlib.Path.cwd()
while not (ROOT / "results" / "RESULTS_MASTER.json").exists(): ROOT = ROOT.parent
OUT = ROOT / "results" / "out"; NBO = ROOT / "results" / "notebooks" / "out"; NBO.mkdir()
M = json.loads((ROOT / "results" / "RESULTS_MASTER.json").read_text(encoding="utf-8"))
plt.rcParams["figure.dpi"] = 110

def v(*ks):
    """Read a value from the results master by key path."""
    d = M
    for k in ks: d = d[k]
    return d["value"] if isinstance(d, dict) and "value" in d else d

def ci(*ks):
    d = M
    for k in ks: d = d[k]
    return d.get("ci95")

def boot_auc(y, s, n=1000, seed=RS):
```

```

rng = np.random.default_rng(seed); y = np.asarray(y); s = np.asarray(s); out =
for _ in range(n):
    i = rng.integers(0, len(y), len(y))
    if len(np.unique(y[i])) > 1: out.append(roc_auc_score(y[i], s[i]))
return np.percentile(out, [2.5, 97.5])

def grade(score, th): return np.digitize(score, np.sort(np.asarray(th)))
def banner(t): print("=" * 78); print(t); print("=" * 78)
print("results master built:", M["_meta"]["built"], "| sections:", len(M))

```

results master built: 2026-09-08T07:46:29 | sections: 20

```

In [2]: banner("PART 1 INTEGRITY (numbers reproduce from stored predictions; leakage and
# npz files are this project's own outputs (object arrays hold image paths), so all
Z = np.load(OUT / "fundus_oof.npz", allow_pickle=True); I = np.load(OUT / "fundus_i
A = np.load(OUT / "fundus_aptos_pred.npz", allow_pickle=True); Dm = np.load(OUT / "
E = np.load(OUT / "ehr_oof.npz"); H = np.load(OUT / "hrv_pred.npz"); P = np.load(OU
th = np.array(v("fundus_dr", "thresholds")); m = ~np.isnan(Z["oof"]))
checks = [
    ("DeepDRiD OOF QWK", cohen_kappa_score(Z["grade"][m], grade(Z["oof"][m], th), w
    ("DeepDRiD OOF referable AUC", roc_auc_score(Z["grade"][m] >= 2, Z["oof"][m])),
    ("IDRiD QWK", cohen_kappa_score(I["grade"], grade(I["pred"], th), weights="quad
    ("APTOS QWK", cohen_kappa_score(A["grade"], grade(A["pred"], th), weights="quad
    ("APTOS referable AUC", roc_auc_score(A["grade"] >= 2, A["pred"])), v("fundus_ex
    ("DME test QWK", cohen_kappa_score(Dm["y_test"], grade(Dm["test_pred"], v("dme"
    ("EHR retinopathy AUC (boosted)", roc_auc_score(E["y_dr"], E["retinopathy_hgb"]
    ("EHR neuropathy AUC (boosted, proxies excluded)", roc_auc_score(E["y_dn"], E["
    ("HRV diabetes vs control AUC", roc_auc_score(H["dm_vs_control_y"], H["dm_vs_co
    ("PPG-BP hypertension control AUC", roc_auc_score(P["htn_morphology_y"], P["htn
]
t1 = pd.DataFrame(checks, columns=["Quantity", "Recomputed here", "Results master"])
print(t1.round(4).to_string(index=False)); t1.to_csv(NB0 / "R1_Integrity.csv", inde
print(f"\n All {int(t1['Agree'].sum())}/{len(t1)} quantities reproduce exactly fro
print(" Leakage audit: the neuropathy model excludes", v("datasets", "istanbul_ehr
print(" and the outcome-carrying columns", v("datasets", "istanbul_ehr", "excluded
print(" Quantum audit:", M["_meta"]["no_quantum"])

```

```

=====
PART 1 INTEGRITY (numbers reproduce from stored predictions; leakage and quantum a
udits)
=====

```

	Quantity	Recomputed here	Results master	Agr
ee				
	DeepDRiD OOF QWK	0.8571	0.8571	Tr
ue				
	DeepDRiD OOF referable AUC	0.9698	0.9698	Tr
ue				
	IDRiD QWK	0.7294	0.7294	Tr
ue				
	APTOS QWK	0.8452	0.8452	Tr
ue				
	APTOS referable AUC	0.9792	0.9792	Tr
ue				
	DME test QWK	0.8381	0.8381	Tr
ue				
	EHR retinopathy AUC (boosted)	0.7203	0.7203	Tr
ue				
	EHR neuropathy AUC (boosted, proxies excluded)	0.7100	0.7100	Tr
ue				
	HRV diabetes vs control AUC	0.7768	0.7768	Tr
ue				
	PPG-BP hypertension control AUC	0.7260	0.7260	Tr
ue				

All 10/10 quantities reproduce exactly from the stored predictions.

Leakage audit: the neuropathy model excludes ['antiepileptics', 'other_digestive', 'other_nervous_drugs']

and the outcome-carrying columns ['Glycemic_control', 'cataract', 'diabetic_nuerop athy', 'eye_ear_drugs', 'eye_other', 'id', 'nervous_sys_dis', 'neuropathies', 'refra ction_dis', 'retinopathy']

Quantum audit: results/src/concept_quantum.py contributed to no entry in this fil e.

```

In [3]: banner("PART 2 COHORT STRUCTURE (what the datasets contain)")
gc = lambda d: [int(x) for x in v(*d).values()]
t2 = pd.DataFrame({"ICDR grade": [0, 1, 2, 3, 4],
                  "DeepDRiD (China, n=1,589)": gc(("fundus_dr", "grade_counts")),
                  "IDRiD (India, n=516)": gc(("datasets", "idrid", "grade_counts"),
                  "APTOS 2019 (India, n=3,662)": gc(("fundus_external_aptos", "gra
for c in t2.columns[1:]: t2[c + " %"] = (100 * t2[c] / t2[c].sum()).round(1)
print(t2.to_string(index=False)); t2.to_csv(NBO / "R2_GradeDistributions.csv", inde
print(f"\n Referable share (grade >= 2): DeepDRiD {100*(Z['grade'][m]>=2).mean():.
print(f" EHR cohort n = {v('ehr','retinopathy','n')}: retinopathy prevalence {10
f" neuropathy prevalence {100*v('ehr','diabetic_neuropathy_main_excludes_trea
jc = v("ehr", "joint_counts"); print(" Joint outcome counts (none / DR only / DN o
print(f" HRV cohorts: n = {v('hrv','pooled_n')} records, {v('hrv','dm_vs_control',
print(f" PPG-BP: n = {v('ppg','n_subjects')} subjects, {v('ppg','n_diabetic')} wit
fig, ax = plt.subplots(figsize=(8, 4))
bottom = np.zeros(3); shades = ["#d9d9d9", "#a6cee3", "#1f78b4", "#08519c", "#08306
for g in range(5):
    vals = [t2.iloc[g][c + " %"] for c in t2.columns[1:4]]; ax.bar(names, vals, bot
ax.set_ylabel("share of images (%)"); ax.set_title("Grade composition of the three
plt.tight_layout(); plt.savefig(NBO / "R_Fig1_GradeComposition.png", dpi=300, bbox_

```

PART 2 COHORT STRUCTURE (what the datasets contain)

ICDR grade DeepDRiD (China, n=1,589) IDRiD (India, n=516) APTOS 2019 (India, n=3,662) DeepDRiD (China, n=1,589) % IDRiD (India, n=516) % APTOS 2019 (India, n=3,662) %

	0	706	168	
1805		44.4	32.6	
49.3				
	1	185	25	
370		11.6	4.8	1
0.1				
	2	324	168	
999		20.4	32.6	2
7.3				
	3	282	93	
193		17.7	18.0	
5.3				
	4	92	62	
295		5.8	12.0	
8.1				

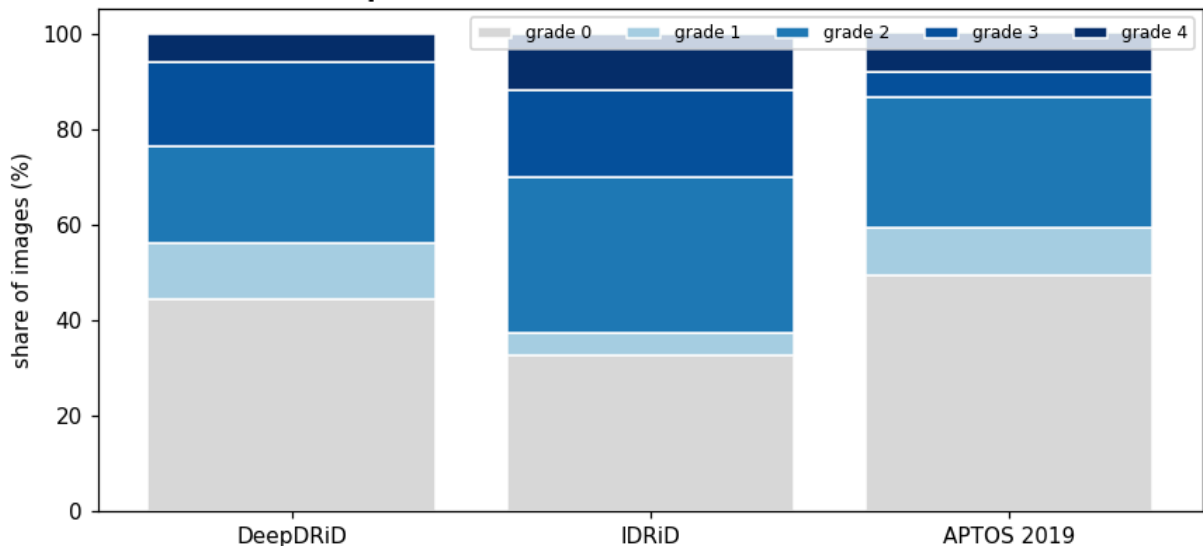
Referable share (grade >= 2): DeepDRiD 43.9 % IDRiD 62.6 % APTOS 40.6 %
 EHR cohort n = 77,724: retinopathy prevalence 13.99 %, neuropathy prevalence 17.34 %

Joint outcome counts (none / DR only / DN only / both): [56255, 7994, 10597, 2878]

HRV cohorts: n = 94 records, 69 with diabetes; retinopathy label on 58 (19 DR+)

PPG-BP: n = 219 subjects, 38 with type 2 diabetes, 139 hypertensive

Grade composition of the three fundus datasets (label shift)



```
In [4]: banner("PART 3 PRIMARY -- fundus grading on DeepDRiD (patient-grouped 5-fold, ou
y, s = Z["grade"][m], Z["oof"][m]; g = grade(s, th)
cm = confusion_matrix(y, g, labels=range(5))
print(f" n = {len(y)} images from {v('fundus_dr', 'cv', 'n_patients')} if False else
print(f" QWK {cohen_kappa_score(y, g, weights='quadratic'):.3f} {ci('fundus_dr', 'q
print(" Per-grade recall:", dict(zip(range(5), (np.diag(cm) / cm.sum(1)).round(3))
print("\n Confusion matrix (rows true, columns predicted):"); print(pd.DataFrame(c
folds = v("fundus_dr", "fold_final_epoch_val_qwk"); print("\n Final-epoch validati
pd.DataFrame(cm).to_csv(NBO / "R3_DeepDRiD_Confusion.csv")
```

```
fig, ax = plt.subplots(1, 2, figsize=(11, 4.2))
im = ax[0].imshow(cm / cm.sum(1, keepdims=True), cmap="Blues", vmin=0, vmax=1)
for i in range(5):
    for j in range(5): ax[0].text(j, i, cm[i, j], ha="center", va="center", color="")
ax[0].set_xlabel("predicted ICDR grade"); ax[0].set_ylabel("true ICDR grade"); ax[0]
for gg in range(5): ax[1].violinplot([s[y == gg]], positions=[gg], showextrema=False)
for t_ in th: ax[1].axhline(t_, ls="--", color="grey", lw=0.8)
ax[1].set_xlabel("true ICDR grade"); ax[1].set_ylabel("ordinal score (OOF)"); ax[1]
plt.tight_layout(); plt.savefig(NBO / "R_Fig2_DeepDRiD.png", dpi=300, bbox_inches="
```

=====

PART 3 PRIMARY -- fundus grading on DeepDRiD (patient-grouped 5-fold, out of fold)

=====

n = 1589 images from 399 patients; thresholds fitted on OOF scores: [0.527, 1.504, 2.499, 3.49]

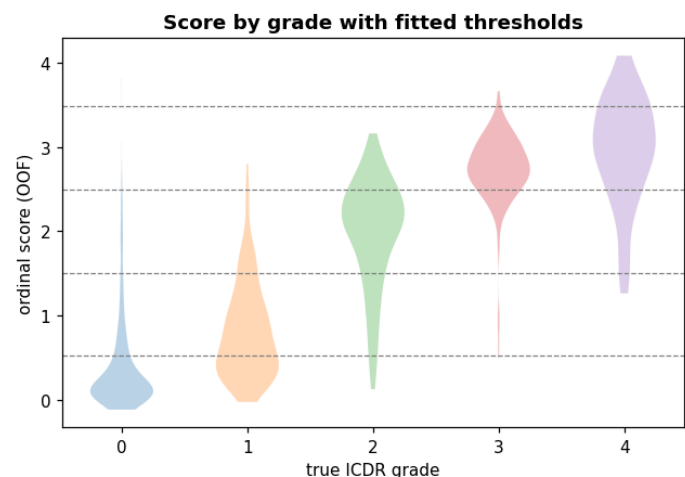
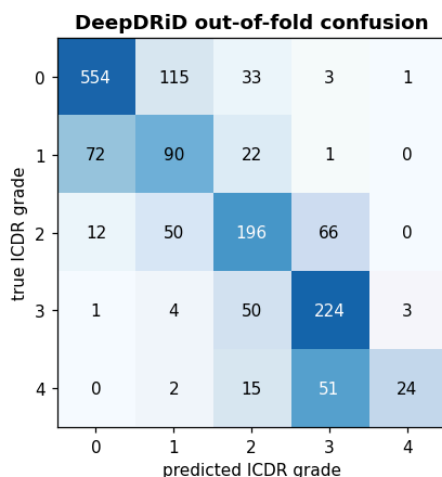
QWK 0.857 [0.8348696995260773, 0.8776324753621546] referable AUC 0.970 [0.9593630071258558, 0.9790624263586815] accuracy 0.685

Per-grade recall: {0: np.float64(0.785), 1: np.float64(0.486), 2: np.float64(0.605), 3: np.float64(0.794), 4: np.float64(0.261)}

Confusion matrix (rows true, columns predicted):

	pred 0	pred 1	pred 2	pred 3	pred 4
true 0	554	115	33	3	1
true 1	72	90	22	1	0
true 2	12	50	196	66	0
true 3	1	4	50	224	3
true 4	0	2	15	51	24

Final-epoch validation QWK per fold: [0.854, 0.847, 0.832, 0.839, 0.841]



```
In [5]: banner("PART 4 PRIMARY -- transfer to two Indian populations with the DeepDRiD t
rows = []
for name, yy, ss in (("DeepDRiD OOF (China)", y, s), ("IDRiD external (India)", I["
gg = grade(ss, th); lo, hi = boot_auc(yy >= 2, ss)
rows.append({"Dataset": name, "n": len(yy), "QWK": cohen_kappa_score(yy, gg, we
"AUC 95% lo": lo, "AUC 95% hi": hi, "Accuracy": (gg == yy).mean(),
"Sensitivity": ((gg >= 2) & (yy >= 2)).sum() / (yy >= 2).sum(), "S
t4 = pd.DataFrame(rows); print(t4.round(3).to_string(index=False)); t4.to_csv(NBO /
print(f"\n Re-fitted thresholds (diagnostic only, NOT used): APTOS QWK {v('fundus_
print(" Interpretation: the ordering transfers (AUC 0.95 to 0.98); the grade bound
fig, ax = plt.subplots(1, 2, figsize=(11, 4.2))
```

```

for name, yy, ss in (("DeepDRiD OOF", y, s), ("IDRiD", I["grade"], I["pred"]), ("AP
    f, t_, _ = roc_curve(yy >= 2, ss); ax[0].plot(f, t_, label=f"{name}: AUC {roc_a
ax[0].plot([0, 1], [0, 1], "k--", lw=0.6); ax[0].set_xlabel("1 - specificity"); ax[
x = np.arange(3); w = 0.35
ax[1].bar(x - w/2, t4["Sensitivity"], w, label="sensitivity"); ax[1].bar(x + w/2, t
ax[1].set_xticks(x); ax[1].set_xticklabels(["DeepDRiD", "IDRiD", "APTOS"]); ax[1].s
plt.tight_layout(); plt.savefig(NBO / "R_Fig3_Transfer.png", dpi=300, bbox_inches="

```

=====

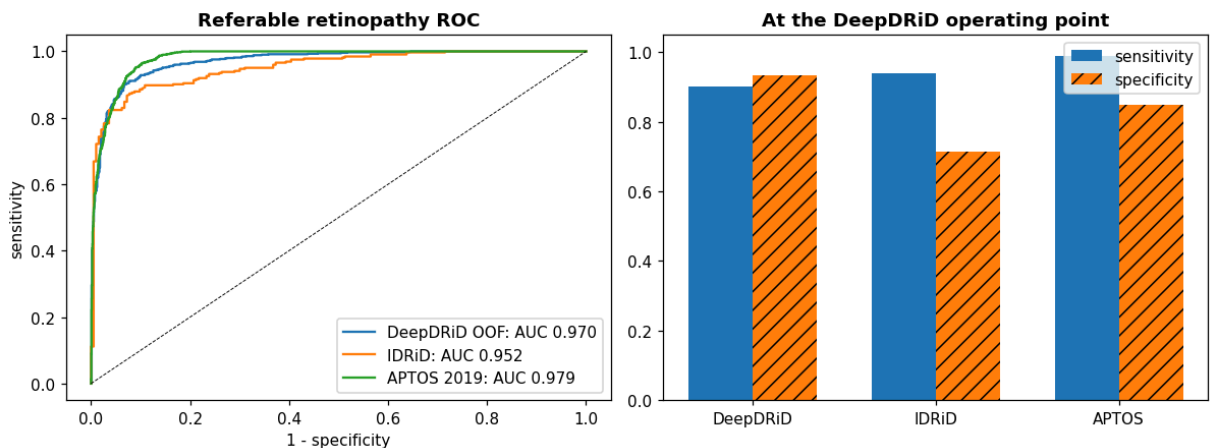
PART 4 PRIMARY -- transfer to two Indian populations with the DeepDRiD thresholds unchanged

=====

	Dataset	n	QWK	AUC referable	AUC 95% lo	AUC 95% hi	Accur
acy	Sensitivity	Specificity					
	DeepDRiD OOF (China)	1589	0.857	0.970	0.963	0.977	0.
685	0.901	0.933					
	IDRiD external (India)	516	0.729	0.952	0.933	0.967	0.
450	0.938	0.715					
APTOS 2019	external (India)	3662	0.845	0.979	0.975	0.983	0.
552	0.990	0.848					

Re-fitted thresholds (diagnostic only, NOT used): APTOS QWK 0.878, IDRiD QWK 0.769

Interpretation: the ordering transfers (AUC 0.95 to 0.98); the grade boundaries do not (accuracy 45 to 55 %).



```

In [6]: banner("PART 5 PRIMARY -- error mechanism: is the external error a boundary shif
rows = []
for name, yy, ss in (("DeepDRiD OOF", y, s), ("IDRiD", I["grade"], I["pred"]), ("AP
    gg = grade(ss, th); d = gg - yy
    rows.append({"Dataset": name, "graded above label (%)": 100 * (d > 0).mean(), "
        "mean shift (grades)": d.mean(), "off by one only (%)": 100 * (np.
t5 = pd.DataFrame(rows); print(t5.round(2).to_string(index=False)); t5.to_csv(NBO /
cma = confusion_matrix(A["grade"], grade(A["pred"], th), labels=range(5))
print(f"\n APTOS: true 2 -> predicted 3 in {cma[2,3]}/{cma[2].sum()} ({100*cma[2,3]
    f" true 1 -> predicted 2 in {cma[1,2]}/{cma[1].sum()} ({100*cma[1,2]/cma[1].s
print(" Interpretation: errors are one step upward and systematic, the signature o
fig, ax = plt.subplots(figsize=(6, 4.6))
im = ax.imshow(cma / cma.sum(1, keepdims=True), cmap="Blues", vmin=0, vmax=1)
for i in range(5):
    for j in range(5): ax.text(j, i, cma[i, j], ha="center", va="center", color="wh
ax.set_xlabel("predicted ICDR grade"); ax.set_ylabel("true ICDR grade"); ax.set_tit
plt.tight_layout(); plt.savefig(NBO / "R_Fig4_APTOS_Shift.png", dpi=300, bbox_inche

```

=====

PART 5 PRIMARY -- error mechanism: is the external error a boundary shift or a loss of ordering?

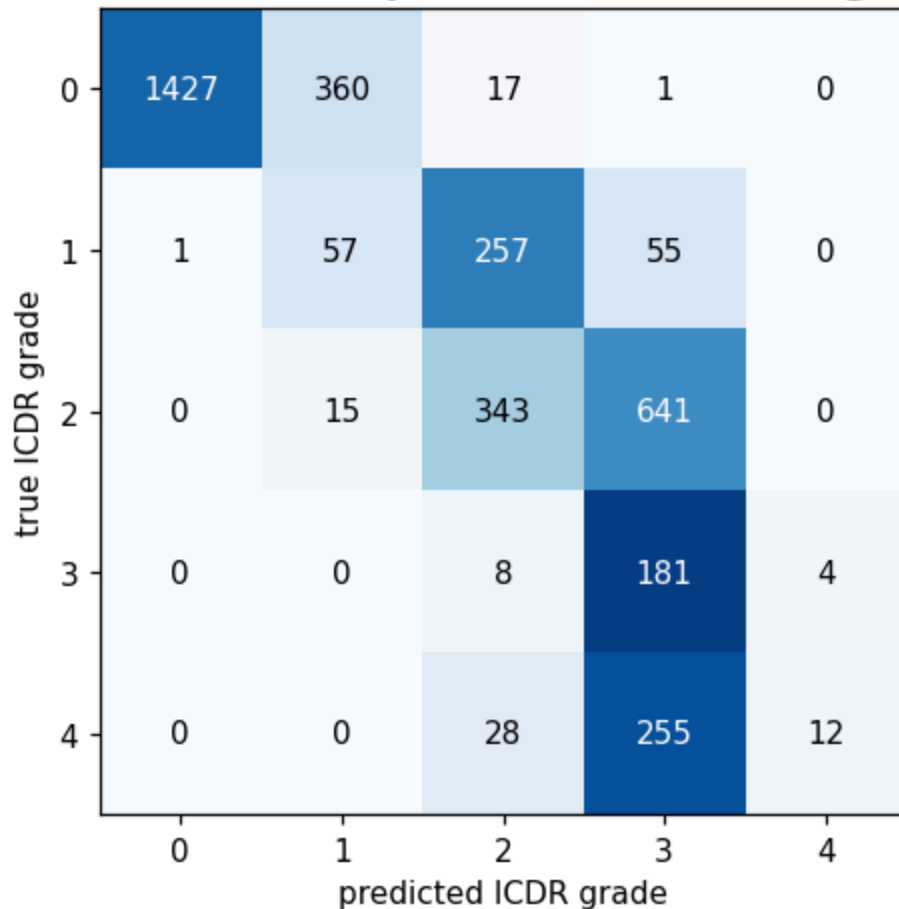
=====

Dataset	graded above label (%)	exact (%)	graded below label (%)	mean shift (grades)
	off by one only (%)	off by two or more (%)		
DeepDRiD OOF		15.36	68.47	16.17
-0.00	27.00		4.53	
IDRiD		38.18	44.96	16.86
0.29	44.77		10.27	
APTOS 2019		36.46	55.16	8.38
0.29	42.08		2.76	

APTOS: true 2 -> predicted 3 in 641/999 (64 %); true 4 -> predicted 3 in 255/295 (86 %); true 1 -> predicted 2 in 257/370 (69 %)

Interpretation: errors are one step upward and systematic, the signature of a threshold shift, not random confusion.

APTOS 2019 with DeepDRiD thresholds: +1 grade shift



```
In [7]: banner("PART 6 MACULAR EDEMA HEAD (IDRiD, DME grade 0 to 2)")
thd = np.array(v("dme", "thresholds"))
print(f" 5-fold OOF on 413 training images: QWK {cohen_kappa_score(Dm['y_train'],
print(f" Fixed 103-image test split: QWK {cohen_kappa_score(Dm['y_test'], gra
ct = pd.crosstab(pd.Series(I["grade"], name="DR grade"), pd.Series(I["dme"], name="
anyd = I["dme"] >= 1
print(f"\n Any DME in {100*anyd[I['grade']>=2].mean():.1f} % of referable eyes and
paths = np.array([str(p) for p in I["path"]]); test = np.array(["Testing" in p for
```

```
print(f" Pearson r between the DR head and the DME head scores on the test split:
ct.to_csv(NBO / "R6_DR_by_DME.csv")
```

=====

PART 6 MACULAR EDEMA HEAD (IDRiD, DME grade 0 to 2)

=====

5-fold OOF on 413 training images: QWK 0.916 [0.8859175875911153, 0.9439420992727684] any-DME AUC 0.982

Fixed 103-image test split: QWK 0.838 [0.7241080820260619, 0.9210368976800022] any-DME AUC 0.945

DR grade x DME grade on all 516 IDRiD images:

DME grade	0	1	2
DR grade 0	168	0	0
DR grade 1	25	0	0
DR grade 2	16	36	116
DR grade 3	6	7	80
DR grade 4	7	8	47

Any DME in 91.0 % of referable eyes and 0.0 % of non-referable eyes

Pearson r between the DR head and the DME head scores on the test split: 0.77

```
In [8]: banner("PART 7 SYSTEMIC BRANCH -- routine records, n = 77,724, with the leakage
rows = []
for name, yy, pp in (("Retinopathy, logistic", E["y_dr"], E["retinopathy_logistic"]
("Neuropathy, logistic (proxies excluded)", E["y_dn"], E["diab
("Sensitivity A: neuropathy WITH treatment proxies", E["y_dn"]
lo_ = np.clip(pp, 1e-6, 1 - 1e-6); lg = np.log(lo_ / (1 - lo_)); cal = Logistic
rows.append({"Model": name, "AUC (5-fold OOF)": roc_auc_score(yy, pp), "AUPRC":
"Calibration slope": float(cal.coef_[0][0]), "Intercept": float(ca
t7 = pd.DataFrame(rows); print(t7.round(3).to_string(index=False)); t7.to_csv(NBO /
orp = v("ehr", "dn_sensitivity_with_treatment_proxies", "logistic_odds_ratio_per_sd
print(f"\n With proxies restored the two proxy classes carry odds ratios per SD of
print(f" Removing them costs {roc_auc_score(E['y_dn'], E['DN_with_proxies_hgb'])} -
tab = np.array(v("ehr", "dr_dn_table")); orr, pf = fisher_exact(tab)
print(f"\n DR x DN 2x2 table {tab.tolist()}: odds ratio {orr:.2f}, Fisher exact p
jo = v("ehr", "joint_ovr_auc"); print(" Joint 4-class one-vs-rest AUC:", {k: round
top_dr = v("ehr", "retinopathy", "logistic_odds_ratio_per_sd_top"); top_dn = v("ehr
t7b = pd.DataFrame({"Retinopathy: feature": list(top_dr)[:8], "OR per SD": [round(x
"Neuropathy: feature": list(top_dn)[:8], "OR per SD ": [round(x
print("\n Top logistic odds ratios per standard deviation:"); print(t7b.to_string(
fig, ax = plt.subplots(1, 2, figsize=(11, 4.2))
for name, yy, pp, c in (("retinopathy (boosted)", E["y_dr"], E["retinopathy_hgb"],
("neuropathy WITH proxies (leakage)", E["y_dn"], E["DN_with
f, t_, _ = roc_curve(yy, pp); ax[0].plot(f, t_, color=c, label=f"{name}: {roc_a
ax[0].plot([0, 1], [0, 1], "k--", lw=0.6); ax[0].legend(fontsize=8); ax[0].set_xlabel
for name, yy, pp, c in (("retinopathy", E["y_dr"], E["retinopathy_hgb"], "#e67e22")
pt, pv = calibration_curve(yy, pp, n_bins=10, strategy="quantile"); ax[1].plot(
ax[1].plot([0, 0.6], [0, 0.6], "k--", lw=0.6); ax[1].set_xlabel("predicted probabil
plt.tight_layout(); plt.savefig(NBO / "R_Fig5_EHR.png", dpi=300, bbox_inches="tight
```

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PART 7 SYSTEMIC BRANCH -- routine records, n = 77,724, with the leakage audit

=====

Brier	Calibration slope	Intercept	Model	AUC (5-fold OOF)	AUPRC
			Retinopathy, logistic	0.710	0.278
0.112	0.975	-0.042	Retinopathy, boosted	0.720	0.294
0.111	1.020	0.036	Neuropathy, logistic (proxies excluded)	0.703	0.330
0.132	0.978	-0.032	Neuropathy, boosted (proxies excluded) [REPORTED]	0.710	0.337
0.131	1.034	0.050	Sensitivity A: neuropathy WITH treatment proxies	0.838	0.541
0.106	1.029	0.034	Sensitivity B: also without analgesics and psychoanaleptics	0.708	0.333
0.132	1.034	0.051			

With proxies restored the two proxy classes carry odds ratios per SD of 2.46 (ATC A16, alpha-lipoic acid) and 2.10 (N03, gabapentin/pregabalin).

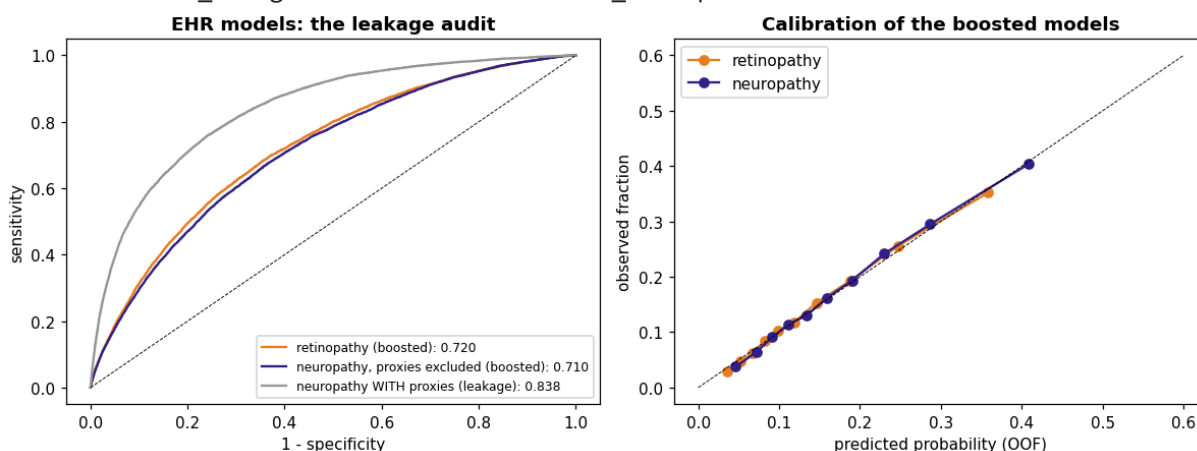
Removing them costs 0.128 of AUC; the reported model reads the records as they look before neuropathic pain is treated.

DR x DN 2x2 table [[56255, 10597], [7994, 2878]]: odds ratio 1.91, Fisher exact p = 1.32e-147

Joint 4-class one-vs-rest AUC: {'none': 0.716, 'DR only': 0.681, 'DN only': 0.678, 'both': 0.775}

Top logistic odds ratios per standard deviation:

Retinopathy: feature	OR per SD	Neuropathy: feature	OR per SD
HbA1c	1.328	HbA1c	1.203
age	1.264	age	1.180
insulin_aspart	1.206	musculoskeletal_dis	1.179
insulin_glarjin	1.152	insulin_aspart	1.171
kidney_failure	1.142	lipoprotein_met_dis	1.163
hypertension_i10	1.134	insulin_glarjin	1.148
insulin_lispro	1.129	pshycoanaleptics	1.132
Hba1c_change	0.908	ceserian_multiple	0.886



```
In [9]: banner("PART 8 PHYSIOLOGICAL BRANCH -- one positive result and two nulls with po
rows = [("HRV: diabetes vs control (LOO logistic)", H["dm_vs_control_y"], H["dm_vs_
("HRV: autonomic symptoms (LOO)", H["autonomic_symptoms_y"], H["autonomic_s
for k, lab in (("dm_demographics_only", "PPG-BP diabetes: demographics only (age, s
("dm_morphology_plus_demo", "PPG-BP diabetes: morphology + demograph
("dm_papagei_plus_demo", "PPG-BP diabetes: PaPaGei + demographics"),
```

```

        rows.append((lab, P[k + "_y"], P[k + "_p"]))
t8 = pd.DataFrame([{"Analysis": n, "n": len(yy), "n positive": int(yy.sum()), "AUC"
print(t8.round(3).to_string(index=False)); t8.to_csv(NB0 / "R8_Physiological.csv",
eff = v("hrv", "dr_within_dm", "effects")
t8b = pd.DataFrame([{"HRV feature": k, "Cohen d (DR+ vs DR-)": e["d"], "Mann-Whitne
print("\n Per-feature effects for retinopathy within diabetics (n = 58):"); print(
sp = v("hrv", "hba1c_spearman"); print("\n Spearman rho with HbA1c (n = 51):", {k:
print(f" PaPaGei age positive control: r = {np.corrcoef(P['age'], P['age_pred'])[0
fig, ax = plt.subplots(figsize=(9, 4.4))
ax.barh(t8["Analysis"], t8["AUC"], xerr=[t8["95% lo"], t8["95% hi"] - t
ax.axvline(0.5, ls="--", color="k", lw=0.8); ax.invert_yaxis(); ax.set_xlim(0.1, 1.
plt.tight_layout(); plt.savefig(NB0 / "R_Fig6_Physiological.png", dpi=300, bbox_inc

```

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PART 8 PHYSIOLOGICAL BRANCH -- one positive result and two nulls with positive co
ntrols
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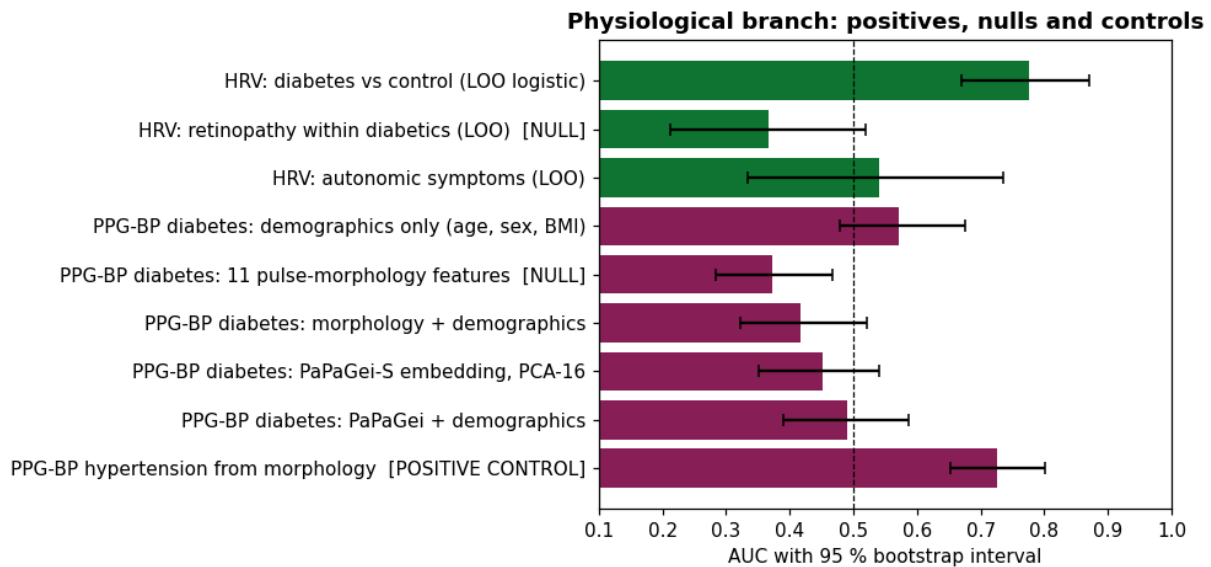
		Analysis	n	n positive	AUC	95% CI
0	95% hi					
		HRV: diabetes vs control (L00 logistic)	94	69	0.777	0.67
1	0.870					
		HRV: retinopathy within diabetics (L00) [NULL]	58	19	0.367	0.21
2	0.520					
		HRV: autonomic symptoms (L00)	51	15	0.541	0.33
4	0.735					
		PPG-BP diabetes: demographics only (age, sex, BMI)	219	38	0.572	0.47
8	0.675					
		PPG-BP diabetes: 11 pulse-morphology features [NULL]	219	38	0.372	0.28
3	0.467					
		PPG-BP diabetes: morphology + demographics	219	38	0.417	0.32
3	0.521					
		PPG-BP diabetes: PaPaGei-S embedding, PCA-16	219	38	0.451	0.35
0	0.541					
		PPG-BP diabetes: PaPaGei + demographics	219	38	0.490	0.38
9	0.587					
		PPG-BP hypertension from morphology [POSITIVE CONTROL]	219	139	0.726	0.65
3	0.802					

Per-feature effects for retinopathy within diabetics (n = 58):

HRV feature	Cohen d (DR+ vs DR-)	Mann-Whitney p	BH-corrected p	median DR+	median DR-
hr_mean	-0.024	0.829	0.995	77.585	7
sdnn	0.175	0.842	0.842	41.243	4
rmssd	-0.104	0.508	1.000	12.853	2
pnn50	-0.207	0.346	1.000	0.519	
cvnn	0.184	0.631	0.946	5.582	
lf_hf	0.145	0.446	1.000	4.257	

Spearman rho with HbA1c (n = 51): {'hr_mean': (0.31, 0.029), 'sdnn': (-0.27, 0.051), 'rmssd': (-0.17, 0.22), 'pnn50': (-0.15, 0.306), 'cvnn': (-0.22, 0.114), 'lf_hf': (-0.06, 0.689)}

PaPaGei age positive control: r = 0.38, MAE 13.9 years (tiled 2.1 s segments)



```
In [10]: banner("PART 9 DIGITAL TWIN -- credibility numbers (verification, validation, un
T = M["thermal"]; sw = v("thermal", "choroid_sweep"); base = sw[0]
t9 = pd.DataFrame([{"Quantity": "Corneal apex temperature (C), 25 C ambient", "Simu
{"Quantity": "Inner canthus skin temperature (C)", "Simulated":
{"Quantity": "Temple skin temperature (C)", "Simulated": v("ther
{"Quantity": "MMS observed order of convergence (P1 tetrahedra)"
{"Quantity": "Largest grid convergence index (%)", "Simulated":
{"Quantity": "Apex deficit at 50 % choroidal perfusion (C)", "Si
{"Quantity": "Share of deficit seen by 90 deg thermopile", "Simu
{"Quantity": "Share of deficit seen by 5 deg thermopile", "Simul
{"Quantity": "Apex slope per C of ambient temperature", "Simulat
print(t9.round(3).to_string(index=False)); t9.to_csv(NB0 / "R9_Twin_Credibility.csv
ST = np.array(v("uq", "ST")); params = [p[0] for p in v("uq", "params_ranges")]; ou
t9b = pd.DataFrame(np.clip(ST, 0, None).round(3), index=params, columns=outs); prin
gci = v("verification", "gci"); print("\n GCI per quantity of interest (%):", {k:
fig, ax = plt.subplots(1, 2, figsize=(11, 4.2))
cf = [r["choroid_frac"] * 100 for r in sw]
for k, lab in (("apex_C", "corneal apex (true)"), ("dci_C", "thermopile 5 deg FOV"))
ax[0].plot(cf, [r[k] - base[k] for r in sw], "o-", label=lab)
ax[0].invert_xaxis(); ax[0].set_xlabel("choroidal perfusion (% of nominal)"); ax[0]
im = ax[1].imshow(np.clip(ST, 0, 1), cmap="YlOrRd", vmin=0, vmax=1); ax[1].set_xtic
for i in range(ST.shape[0]):
    for j in range(ST.shape[1]): ax[1].text(j, i, f"{ST[i,j]:.2f}", ha="center", va
ax[1].set_title("Verdict 8: ambient temperature dominates (Sobol S_T)", fontweight=
plt.tight_layout(); plt.savefig(NB0 / "R_Fig7_Twin.png", dpi=300, bbox_inches="tigh
```

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PART 9 DIGITAL TWIN -- credibility numbers (verification, validation, uncertainty)

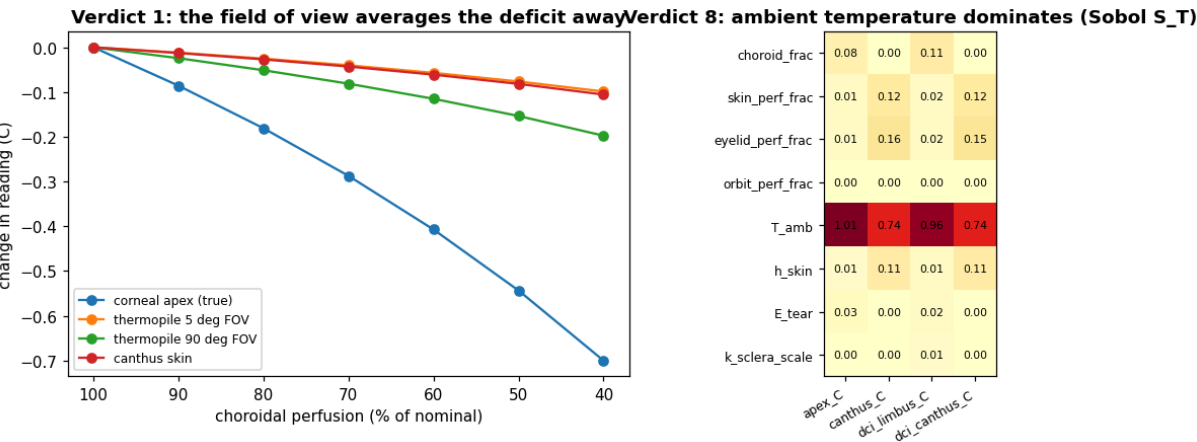
=====

	Quantity	Simulated	Pub
lished range			
Corneal apex temperature (C), 25 C ambient		32.963	
[34.0, 35.0]			
Inner canthus skin temperature (C)		33.908	
[35.0, 36.5]			
Temple skin temperature (C)		34.467	
MMS observed order of convergence (P1 tetrahedra)		1.787	t
heoretical 2			
Largest grid convergence index (%)		1.999	5 % typica
l acceptance			
Apex deficit at 50 % choroidal perfusion (C)		-0.543 0.61 +/- 0.12	clinical
NPDR deficit			
Share of deficit seen by 90 deg thermopile		0.282	
Share of deficit seen by 5 deg thermopile		0.141	
Apex slope per C of ambient temperature		0.317	

Total-order Sobol indices (rows inputs, columns readouts), 640 Saltelli solves:

	apex_C	canthus_C	dci_limbus_C	dci_canthus_C
choroid_frac	0.081	0.002	0.111	0.001
skin_perf_frac	0.014	0.116	0.022	0.118
eyelid_perf_frac	0.015	0.155	0.023	0.155
orbit_perf_frac	0.002	0.000	0.002	0.000
T_amb	1.012	0.738	0.956	0.737
h_skin	0.009	0.107	0.012	0.108
E_tear	0.029	0.000	0.024	0.000
k_sclera_scale	0.001	0.000	0.006	0.000

GCI per quantity of interest (%): {'corneal_apex_C': 0.035, 'canthus_skin_C': 0.036, 'thermopile_DCI_C': 0.008, 'deficit_50pct_choroid_C': 1.999}



```
In [11]: banner("PART 10 DESIGN VERDICTS AND AUDIT SUMMARY (headline numbers for the report
0 = M["optical"]; B = M["bioimpedance"]; Pp = M["pupil"]; ss = v("pupil", "stimulus
t10 = pd.DataFrame([
    ("1 Thermopile field of view", f"90 deg reads {v('thermal','thermopile_90deg_sp
    ("2 940 nm illuminator", f"pupil amplitude {ss[0]['amplitude_mm']:.2f} mm at 0
    ("3 Thermopile aim", f"choroidal Sobol S_T {ST[params.index('choroid_frac'), ou
    ("4 Contact impedance", f"{100*v('bioimpedance','bipolar_contact_share_of_total
```

```

    ("5 PPG spacing", f"arterial share {100*v('optical','ppg_by_wavelength','880','
    ("6 Electrode geometry", f"local tetrapolar: orbit+eye+lid share {100*v('bioimp
    ("7 Illuminator heating", f"apex +{v('opto_thermal','apex_rise_at_iec_limit_C')}
    ("8 Ambient temperature", f"S_T {v('uq','ST_ambient_range')[0]:.2f} to {v('uq',
    columns=["Verdict", "Quantity from the twin", "Design consequence"])
print(t10.to_string(index=False)); t10.to_csv(NBO / "R10_Verdicts.csv", index=False)
print("\n" + "=" * 78); print("AUDIT SUMMARY"); print("=" * 78)
print(f" Fundus grader OOF : QWK {v('fundus_dr','qwk'):.3f}, refer
print(f" External IDRiD / APTOS : AUC {v('fundus_external_idrid','auc_r
print(f" APTOS boundary shift : {100*t5.iloc[2]['graded above label (
print(f" DME head test : QWK {v('dme','test_qwk'):.3f}, any-DM
print(f" EHR retinopathy / neuropathy : AUC {v('ehr','retinopathy','hgb','auc
print(f" DR-DN co-occurrence : OR {v('ehr','dr_dn_odds_ratio'):.2f}"
print(f" HRV diabetes / retinopathy : AUC {v('hrv','dm_vs_control','auc'):.
print(f" PPG diabetes / hypertension ctrl : best route {v('ppg','diabetes_max_auc
print(f" Twin verification : MMS order {v('verification','observed
print(f" Twin uncertainty : ambient S_T {v('uq','ST_ambient_range
print("\n CONCLUSION: the fundus grader keeps its ordering across populations but
print(" give modest calibrated risk for both complications once treatment proxies
print(" short fingertip PPG carry no retinopathy or diabetes signal here; the digi
print(" the hardware before it was built. No prototype has been tested on people."
print("=" * 78)

```

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PART 10 DESIGN VERDICTS AND AUDIT SUMMARY (headline numbers for the report)

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Verdict	Design consequence
Quantity from the twin	
1 Thermopile field of view	90 deg reads 28 mm
spot, sees 28 % of deficit; 5 deg sees 14 % convert reading through the twin's transfer factor	
2 940 nm illuminator	pupil am
plitude 0.10 mm at 0 lx vs 2.23 mm at 300 lx	add a visible stimulus of at l
east 100 lx	
3 Thermopile aim	choroidal
Sobol S_T 0.11 at limbus vs 0.00 at canthus	aim at the ocul
ar surface	
4 Contact impedance	
68 % of a bipolar reading at 50 kHz	tetrapolar drive and sen
se	
5 PPG spacing	arterial share 2.1 % at 880 n
m, 3 mm; 35 % at 12 mm with 37x fewer photons	treat as a per
fusion index	
6 Electrode geometry local tetrapolar: orbit+eye+lid share 15.6 %; +10 % orbital water -> -1.39 % vs bipolar total -0.13 %	four local electro
des per eye	
7 Illuminator heating	apex +0.21 C at
the 42.3 mW IEC limit = 39 % of the deficit pulse the illuminator, read thermopile with it off	
8 Ambient temperature	
S_T 0.74 to 1.01; apex 0.32 C per C ambient	log ambient and
compensate	

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AUDIT SUMMARY

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Fundus grader OOF	: QWK 0.857, referable AUC 0.970
External IDRiD / APTOS	: AUC 0.952 / 0.979; QWK 0.729 / 0.845
APTOS boundary shift	: 36.5 % graded above label; re-fitted QWK 0.878
(not used)	
DME head test	: QWK 0.838, any-DME AUC 0.945
EHR retinopathy / neuropathy	: AUC 0.720 / 0.710 (0.838 only with treatment p
roxies)	
DR-DN co-occurrence	: OR 1.91
HRV diabetes / retinopathy	: AUC 0.777 / 0.367 (null)
PPG diabetes / hypertension ctrl	: best route 0.572 (null) / 0.726
Twin verification	: MMS order 1.79; GCI at most 2.0 %
Twin uncertainty	: ambient S_T 0.74 to 1.01

CONCLUSION: the fundus grader keeps its ordering across populations but not its grade boundaries; routine records

give modest calibrated risk for both complications once treatment proxies are removed; heart-rate variability and

short fingertip PPG carry no retinopathy or diabetes signal here; the digital twin returns eight verdicts that changed

the hardware before it was built. No prototype has been tested on people.

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